Learning-Enhanced Personnel Scheduling for Flexible Multi-Skilled Workforces
Abridged Literature Review

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1 Introduction and Motivation

Personnel scheduling is central to retail, hospitality, healthcare, and other service settings where demand varies and employees differ in skills, contracts, and preferences. Although the field has been studied extensively in operations research, real-world adoption remains limited because classical models capture feasibility and cost more effectively than tacit human preferences and local operational nuance [7, 16, 12]. This abridged review examines 17 papers from 2004 to 2025. Using recent survey papers to frame the field and Xue et al. (2025) to anchor its frontier, it examines how flexible, multi-skilled, learning-enhanced scheduling responds to that adoption gap [17].

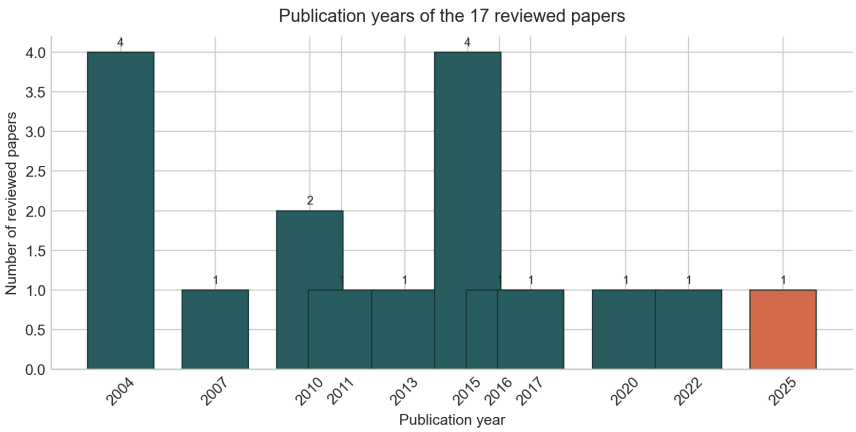

Figure 1: Publication years of the 17 reviewed papers. The concentration after 2010 suggests growing interest in flexible, skill-aware, and implementation-oriented personnel scheduling.

1.1 Research Field Challenges

Four challenges dominate. First, *scalability* worsens when flexible shifts enlarge the search space combinatorially [17]. Second, *heterogeneity* matters because skills, contracts, and availability create interdependent assignment constraints that homogeneous formulations cannot capture well [4, 10]. Third, *demand uncertainty* creates mismatch between staffing plans and realised service needs in service environments [5, 11]. Fourth, a persistent *research-practice gap* means strong objective values do not guarantee adoption if schedules violate tacit rules or fairness expectations [12]. Personnel scheduling is therefore a decision-support problem as well as an assignment problem.

2 Methodology

2.1 Search Strategy

The search started from recent survey papers by Van den Bergh et al. (2013), De Bruecker et al. (2015), and Defraeye and Van Nieuwenhuyse (2016), then used Xue et al. (2025) as a frontier anchor in ConnectedPapers and backward citation chasing. Additional papers were located through ScienceDirect, SpringerLink, IEEE Xplore, and Google Scholar. Keywords included "personnel scheduling", "staff rostering", "flexible shifts", "multi-skilled workforce scheduling", and "preference-aware" or "learning-enhanced" optimisation. The final set retained papers directly relevant to flexibility, heterogeneity, implementation, or data-assisted optimisation.

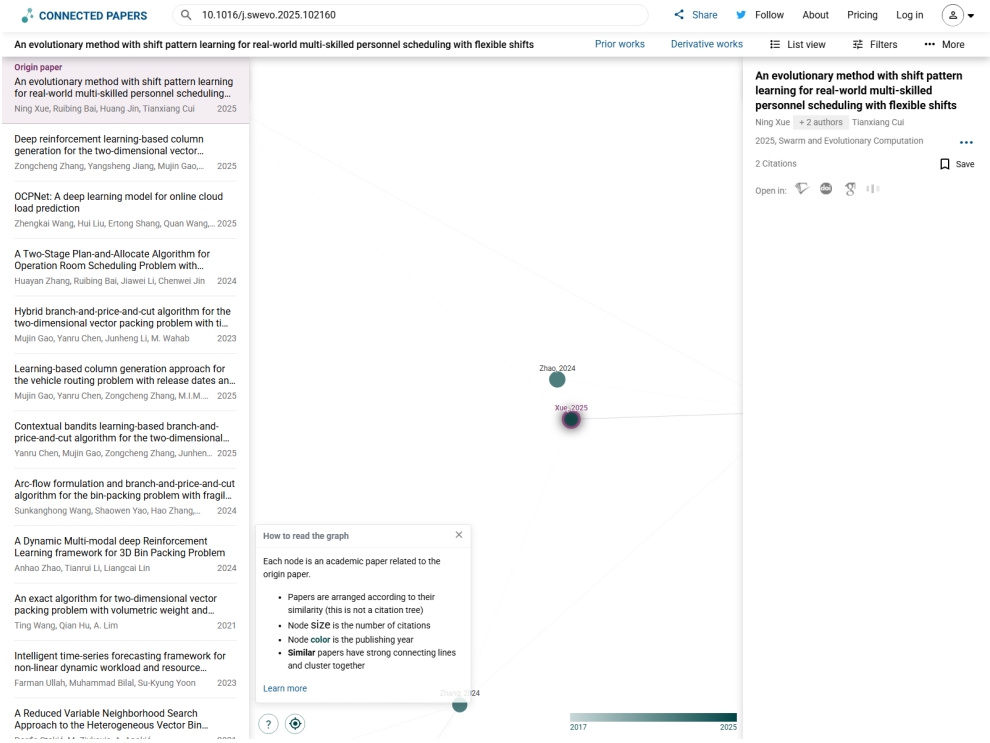

Figure 2: ConnectedPapers graph illustrating citation-based expansion from the frontier anchor paper by Xue et al. (2025), after survey-led scoping.

2.2 Survey Scope

The review includes peer-reviewed papers published between 2004 and 2025 with identifiable DOIs and a clear connection to personnel scheduling, workforce scheduling, or staff rostering. Survey papers were retained when they define the field or classify major modelling choices. Papers were excluded if they focused on routing without workforce scheduling, general HR policy, or lacked a usable DOI. This scope keeps the review close to learning-enhanced scheduling for flexible, multi-skilled workforces.

2.3 Article Statistical Analysis

The search yielded 42 candidate papers. After removing 21 papers for relevance or DOI issues, 26 papers were screened at full-text level and 17 were retained.

Table 1: Article screening summary

Stage	Count
Initial survey papers	3
Frontier anchor paper	1
ConnectedPapers and database candidates	42
Removed after relevance / DOI screening	21
Full-text papers screened	26
Final papers included	17

3 Classification of Literature

The papers are organised by their main contribution: defining the field, addressing implementation and preferences, structuring flexible-shift design, or improving optimisation with hybrid and learning-enhanced methods.

Table 2: Classification scheme for the reviewed literature

Category	Main focus	Papers
A. Surveys and field framing	Field definitions, skill planning, demand uncertainty	Ernst et al. (2004); Van den Bergh et al. (2013); De Bruecker et al. (2015); Defraeye and Van Nieuwenhuyse (2016)
B. Preference and implementation	Human preferences, deployment, industry cases	Gordon and Erkut (2004); Topaloglu and Ozkarahan (2004); Henao et al. (2015); Petrovic (2017)
C. Shift design and decomposition	Shift construction, part-time staffing, uncertainty	Di Gaspero et al. (2007); Hojati and Patil (2011); Parisio and Jones (2015); Prot et al. (2015)
D. Evolutionary and learning-enhanced methods	Evolutionary baselines, matheuristics, ML-guided search	Aickelin and Dowsland (2004); Bai et al. (2010); Çakırgil et al. (2020); Sun et al. (2022); Xue et al. (2025)

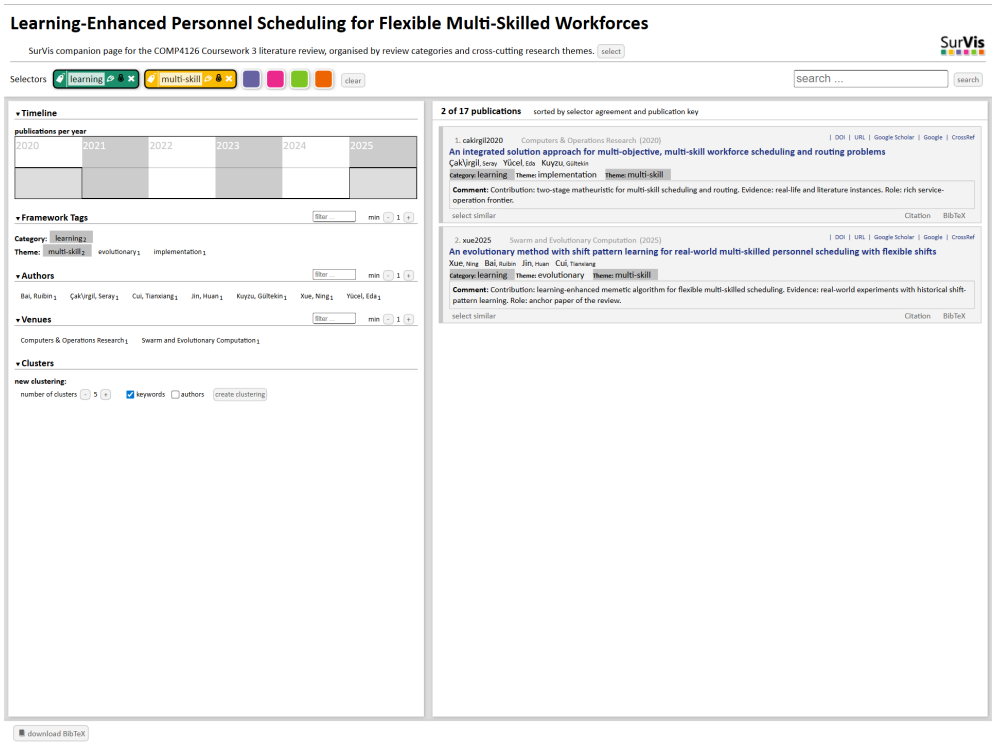

Figure 3: SurVis visualisation of the reviewed literature, showing category-and-theme tags, author distribution, and filtered paper subsets that support the classification framework.

SurVis public URL:
https://lmy12367.github.io/learning-enhanced-scheduling-survis/

3.1 Category A: Surveys and Field Framing

Ernst et al. (2004). This foundational review synthesises applications, models, and solution methods across staff scheduling domains. Its evidence is comparative synthesis rather than experimentation, but it remains a baseline map of objectives and constraints [7].

Van den Bergh et al. (2013). This review classifies personnel scheduling literature by problem setting and technical features through a structured survey. The resulting taxonomy is useful evidence of which feature combinations have been explored and which remain sparse [16].

De Bruecker et al. (2015). This state-of-the-art review focuses on workforce planning with skill considerations. Its synthesis shows that multi-skilled planning changes feasible assignment structure rather than adding a minor side constraint [4].

Defraeye and Van Nieuwenhuyse (2016). This literature review examines staffing and scheduling under nonstationary demand in service systems. The survey evidence shows that fluctuating demand is a central modelling driver rather than a peripheral

complication [5].

These surveys map the field well, but they say less about how preferences and deployment realities enter operational models.

3.2 Category B: Preference and Implementation

Gordon and Erkut (2004). Gordon and Erkut present a spreadsheet decision-support tool for volunteer scheduling at the Edmonton Folk Festival. The evidence is a real implementation case showing that usability and preference handling can matter more than marginal optimisation gains [8].

Topaloglu and Ozkarahan (2004). This paper formulates tour scheduling as implicit goal programming with employee preferences. Computational results show that integrated preference handling can generate feasible schedules without a separate assignment stage [15].

Henao et al. (2015). Henao et al. study multi-skilling in a retail case through a mixed-integer model for training and assignment decisions. Case results suggest that selective, rather than universal, multi-skilling improves schedule efficiency [9].

Petrovic (2017). Petrovic analyses why personnel scheduling research still has limited uptake in practice. The evidence comes from software classification and implementation experience, showing that deployment needs differ from standard benchmark objectives [12].

Compared with survey papers, these studies provide stronger practice evidence, but mostly through isolated case settings rather than transferable validation.

3.3 Category C: Shift Design and Decomposition

Di Gaspero et al. (2007). This paper studies the minimum shift design problem using network-flow structure plus greedy construction and local search. Computational comparisons against an earlier implementation show clear quality gains [6].

Hojati and Patil (2011). Hojati and Patil schedule heterogeneous part-time service employees with an integer linear programming-based heuristic. Their experiments show why mixed contracts and skills materially change the assignment problem [10].

Parisio and Jones (2015). Parisio and Jones model retail employee scheduling as two-stage stochastic programming under uncertain demand. Scenario-based results show that uncertainty changes staffing decisions rather than merely adding robustness penalties [11].

Prot et al. (2015). Prot et al. propose a two-phase method for shift design and personnel task scheduling with an equity objective. Computational evaluation shows that iterative coordination between design and assignment can preserve fairness without collapsing into a naive sequential workflow [13].

Category C improves realism through decomposition and uncertainty handling, but usually sacrifices global integration.

3.4 Category D: Evolutionary and Learning-Enhanced Methods

Aickelin and Dowsland (2004). This paper develops an indirect genetic algorithm for nurse scheduling using permutation encoding and a heuristic decoder. Experiments on live hospital data show competitive performance against tabu search and establish a durable evolutionary baseline [1].

Bai et al. (2010). Bai et al. combine stochastic-ranking genetic search with a simulated-annealing hyper-heuristic for nurse rostering. Comparative experiments show better feasible and high-quality solutions than earlier evolutionary baselines [2].

Çakırgil et al. (2020). This study examines multi-objective, multi-skill workforce scheduling and routing for field service operations. Validation on real-life and literature instances shows that a two-stage matheuristic scales to richer service settings than classical rostering models [3].

Sun et al. (2022). Sun et al. propose ML-ACO, in which a learned predictor biases ant colony optimisation toward promising solution components. Comparative experiments show that learned guidance can strengthen metaheuristic search, although the evidence comes from generic rather than personnel-scheduling instances; it is therefore included as methodological, rather than direct domain, evidence [14].

Xue et al. (2025). Xue et al. address flexible-shift scheduling for multi-skilled part-time staff with a learning-enhanced memetic algorithm. Real-world experiments show that decision-tree learning over historical shift patterns can improve search guidance in a practice-oriented flexible-shift setting, making this the strongest direct evidence for the review's target problem [17].

Category D offers the strongest computational evidence, but most methods still rely on handcrafted encodings and benchmark-style evaluation rather than deployment studies.

4 Discussion

A key insight from the reviewed literature is that the main bottleneck is no longer pure computational optimisation but model realism. Early work emphasised how to solve difficult schedules, whereas recent studies argue that deployment fails when models omit tacit preferences, flexibility, or contextual constraints [12, 17]. There is therefore a clear tension between realism and tractability: flexible shifts, heterogeneous skills, and uncertain demand improve fidelity, but they also push researchers toward decomposition, heuristics, or matheuristics [6, 11, 13]. Evidence strength also varies across categories: survey papers synthesise broad literatures, implementation papers often rely on single organisational cases, and learning-enhanced papers usually report computational comparisons rather than long-horizon field deployment. The reviewed field can therefore be interpreted along two axes: model realism, ranging from fixed and homogeneous formulations to flexible and preference-aware scheduling, and solution intelligence, ranging from exact and decomposition-based optimisation to learning-enhanced search. The SurVis visualisation in Figure 3 reinforces this pattern by showing that only a small frontier of papers lies at the overlap of the learning and multi-skilled themes, while implementation-oriented studies remain concentrated outside that frontier. No reviewed study fully integrates flexible shift design, multi-skilled workforce modelling, data-driven preference learning, and real-world deployment validation in one tested framework. Public benchmarks remain limited, and learned components are rarely evaluated for transferability or bias. Figure 4 summarises this review's synthesis of the field.

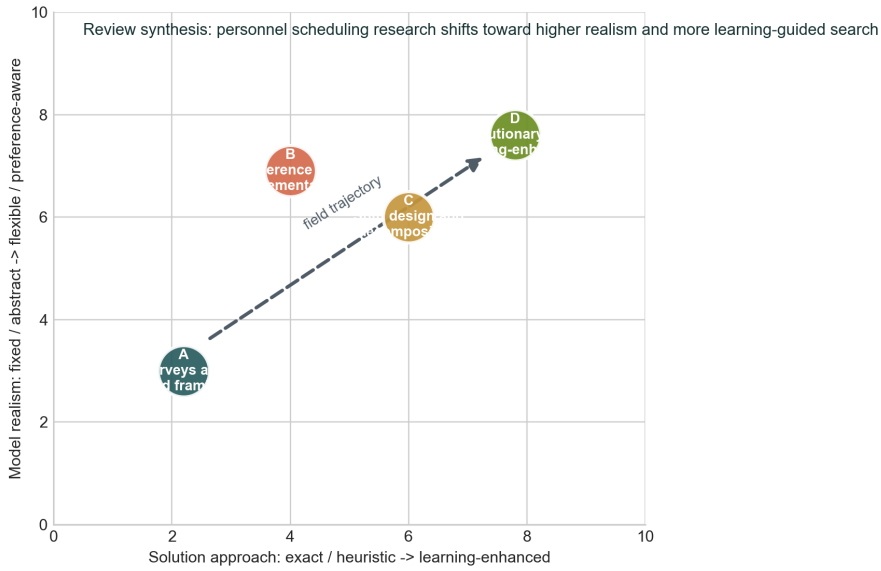

Figure 4: Conceptual framework synthesised from the review. Positions are qualitative category-level synthesis scores: the x-axis reflects solution intelligence from exact/decomposition-based optimisation to learning-enhanced search, and the y-axis reflects model realism from fixed homogeneous settings to flexible, skill-aware, preference-aware formulations.

5 Conclusion

Personnel scheduling research is moving from generic rostering toward flexible, skill-aware decision support, but the next step is interpretable learning-enhanced models validated under real demand volatility and workforce heterogeneity.

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